香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

General Chemistry CHM1001

Lecture 5

Bonding Theories (Chapter9) and Gases (Chapter 10)

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Notice of Class Cancellation

Sep. 25th(Thu)

Online make-up class might be provided before the midterm exam

Recap

Dipoles

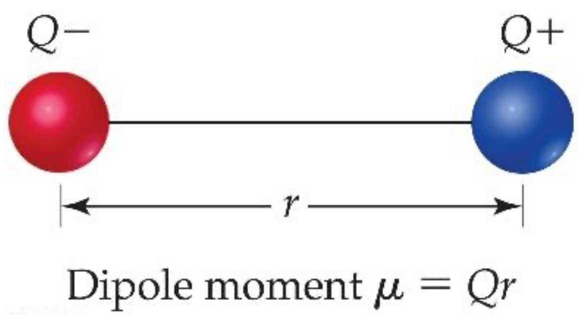
- When two equal, but opposite, charges are separated by a distance, a **dipole** forms.
- A **dipole moment**, μ, produced by two equal but opposite charges separated by a distance, *r*, is calculated:

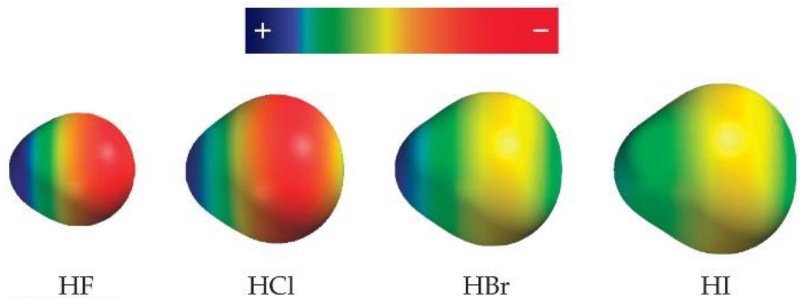
Vector

particles

$$\mu = Qr$$

- It is measured in debyes (D).



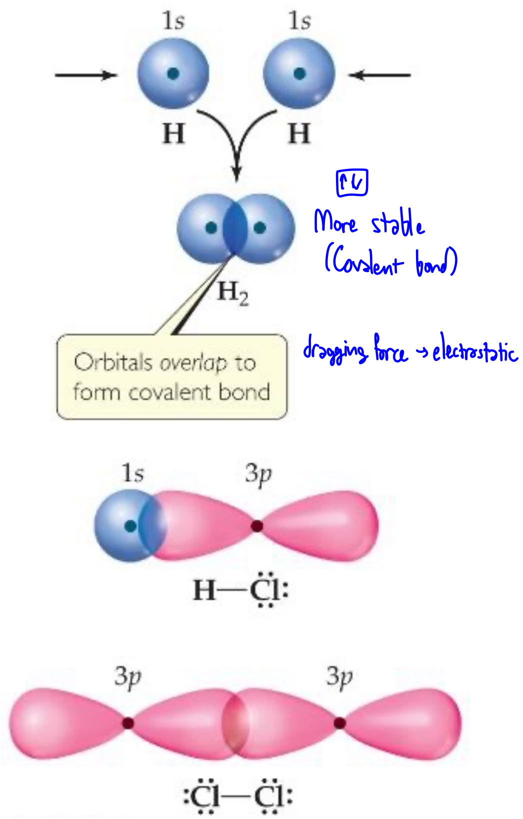


EN สูงสุด

e^- density สูงสุด (ดูได้เยอะ)

Recap

Valence-Bond Theory



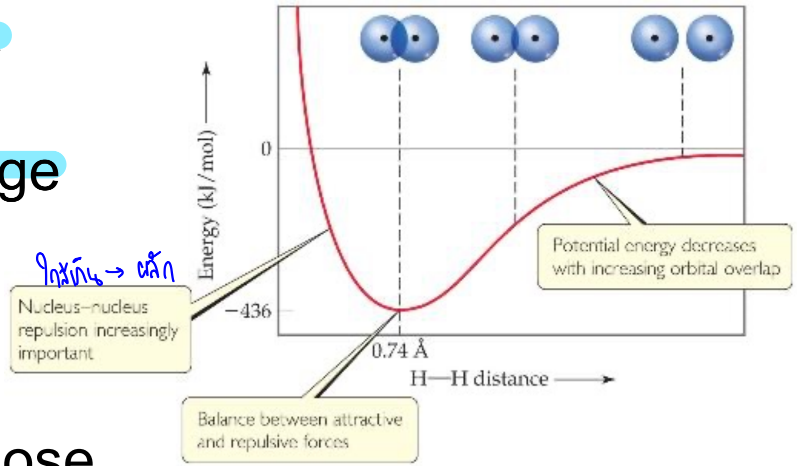
- In Valence-Bond Theory, electrons of two atoms begin to occupy the same space.
- This is called “overlap” of orbitals.
- The sharing of space between two electrons of opposite spin results in a covalent bond.

Recap

Overlap and Bonding

- Increased overlap brings the electrons and nuclei closer together until a balance is reached between the like charge repulsions and the electron-nucleus attraction.
- Atoms can't get too close because the internuclear repulsions get too great.

ใกล้กัน → ผลัก



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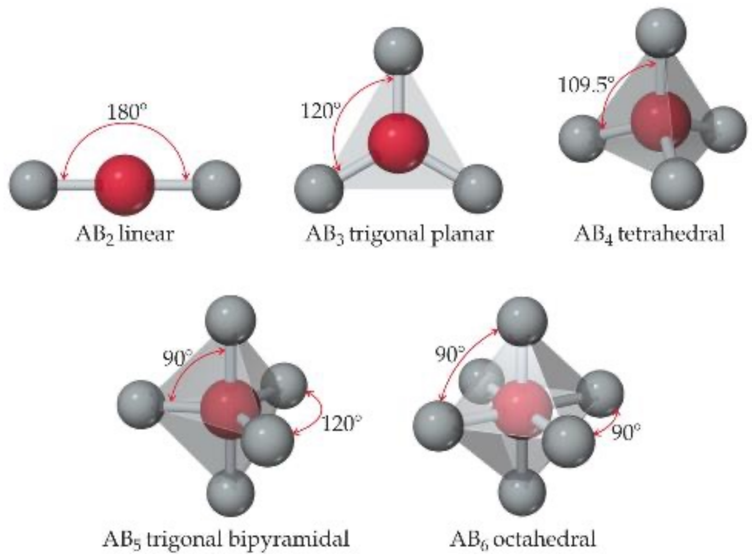
Recap

What Determines the Shape of a Molecule?

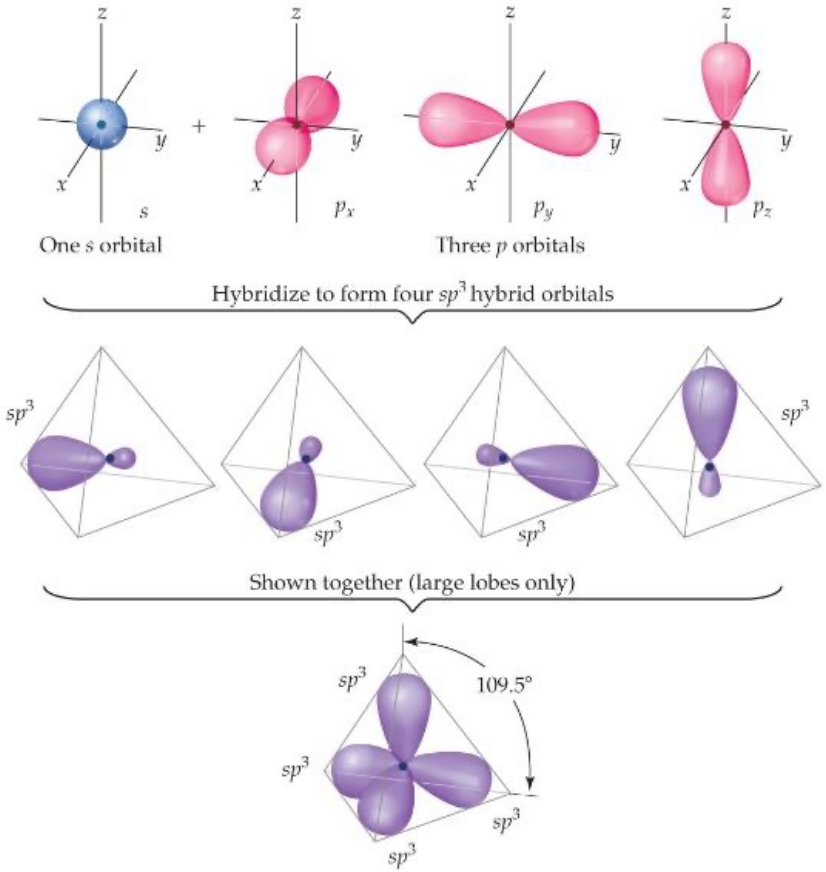
- Simply put, electron pairs, whether they be bonding or nonbonding, repel each other.
- By assuming the electron pairs are placed as far as possible from each other, we can predict the shape of the molecule.
- This is the Valence-Shell Electron-Pair Repulsion (VSEPR) model.



Recap

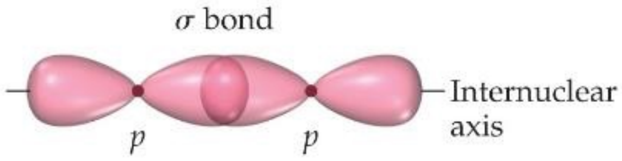
Carbon: sp^3 Hybridization

With carbon, we get four degenerate sp^3 orbitals.



Recap

Sigma (σ) and Pi (π) Bonds



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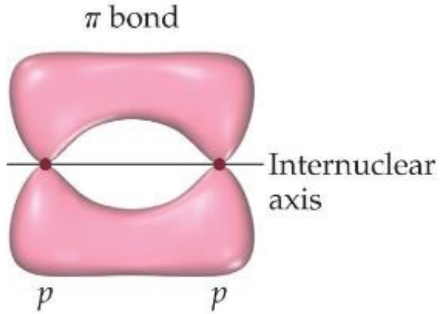
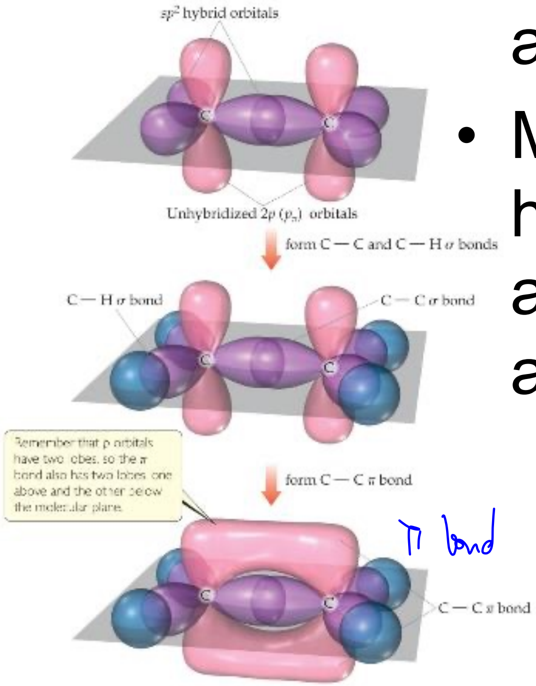


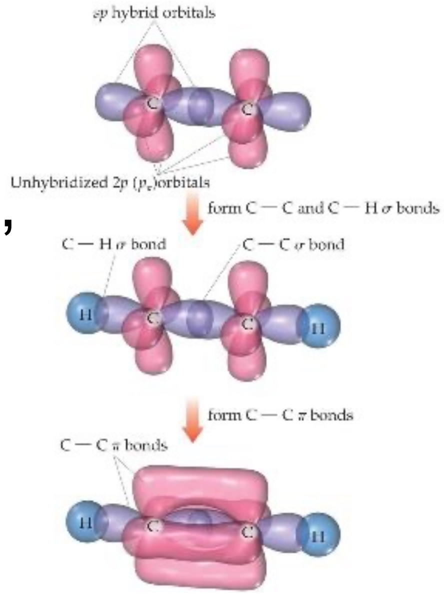
- Sigma bonds are characterized by
  - head-to-head overlap.
  - cylindrical symmetry of electron density about the internuclear axis.
- Pi bonds are characterized by
  - side-to-side overlap.
  - electron density above and below the internuclear axis.

Recap

Bonding in Molecules

- Single bonds are always σ-bonds.
- Multiple bonds have one σ-bond, all other bonds are π-bonds.





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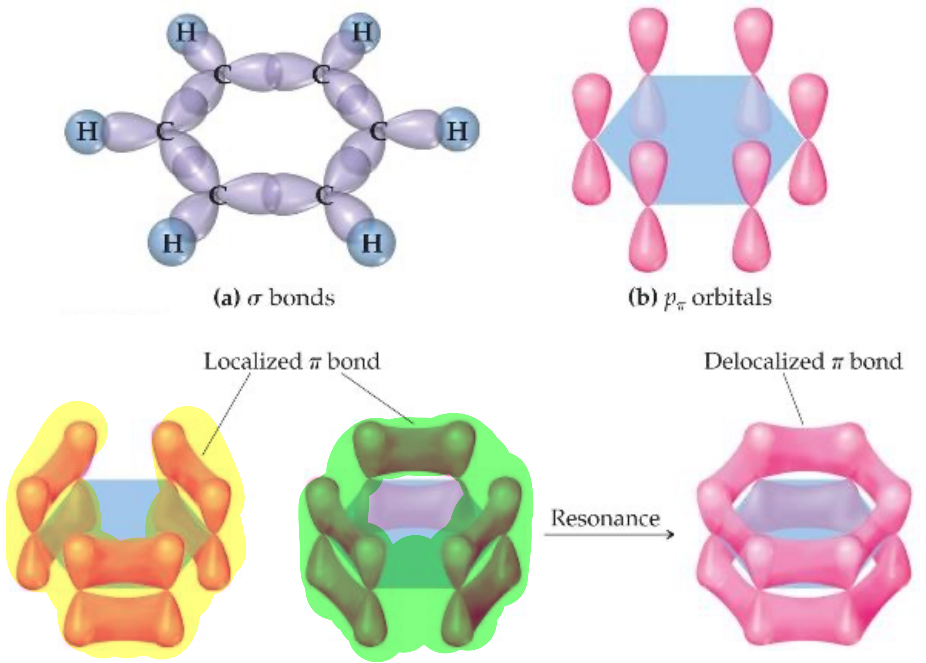
Recap

Benzene

The organic molecule benzene (C_6H_6) has six σ-bonds and a *p* orbital on each C atom, which form delocalized bonds using one electron from each *p* orbital.



MO Theory

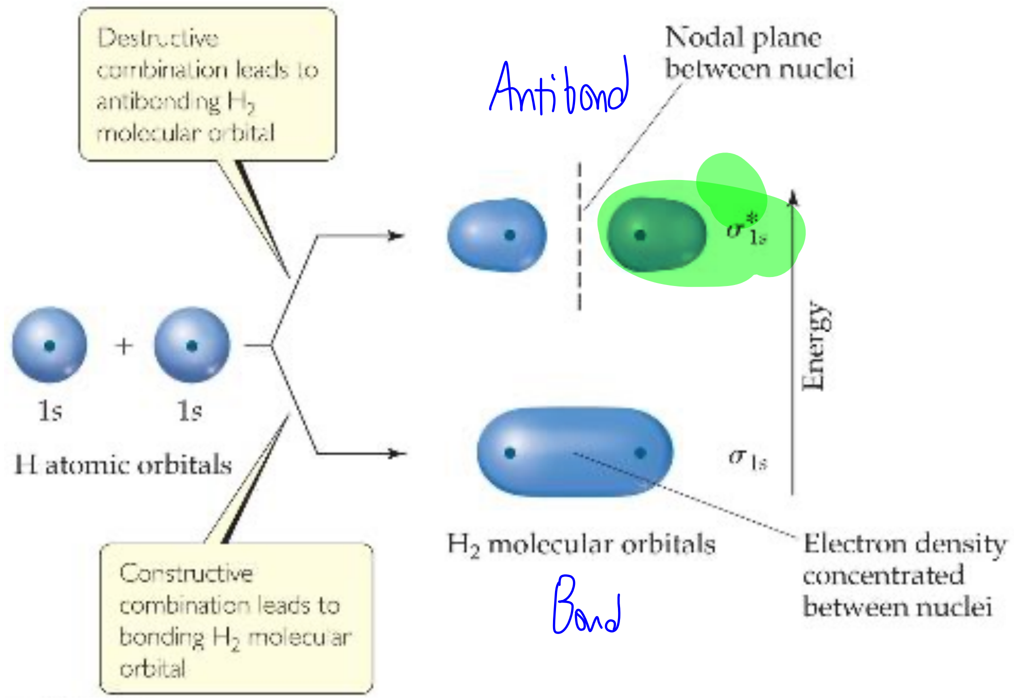
- They differ from atomic orbitals because they represent the entire molecule, not a single atom.
- Whenever two atomic orbitals overlap, two molecular orbitals are formed: one bonding, one antibonding.
- **Bonding orbitals** are constructive combinations of atomic orbitals.
- **Antibonding orbitals** are destructive combinations of atomic orbitals. They have a new feature unseen before: A **nodal plane** occurs where electron density equals zero.

Molecular Orbital (MO) Theory

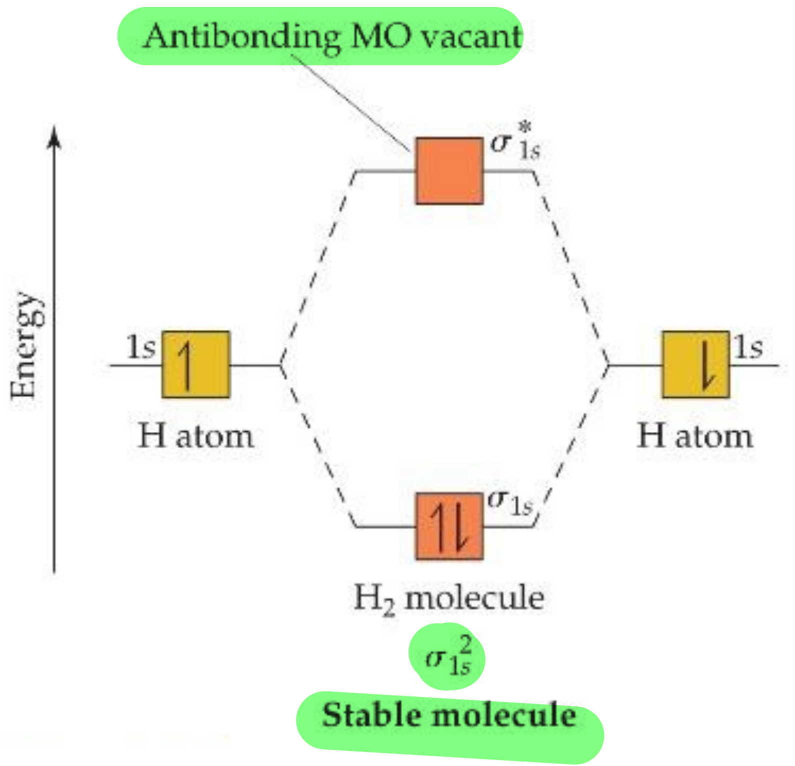
- Wave properties are used to describe the energy of the electrons in a molecule.
- **Molecular orbitals** have many characteristics like atomic orbitals:
  - maximum of two electrons per orbital
  - Electrons in the same orbital have opposite spin.
  - Definite energy of orbital
  - Can visualize electron density by a contour diagram

Molecular Orbital (MO) Theory

Whenever there is direct overlap of orbitals, forming a bonding and an antibonding orbital, they are called **sigma (σ) molecular orbitals**. The antibonding orbital is distinguished with an asterisk as σ^*. Here is an example for the formation of a hydrogen molecule from two atoms.



MO Diagram

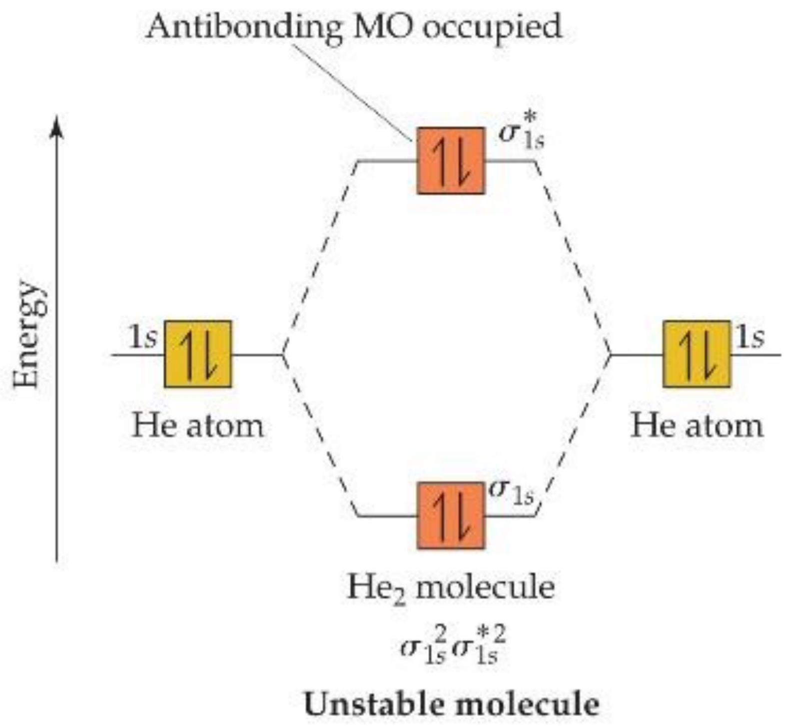


- An **energy-level diagram**, or **MO diagram** shows how orbitals from atoms combine to give the molecule.
- In H_2 the two electrons go into the bonding molecular orbital (lower in energy).
- **Bond order** = ½(# of bonding electrons – # of antibonding electrons) = ½(2 – 0) = 1 bond

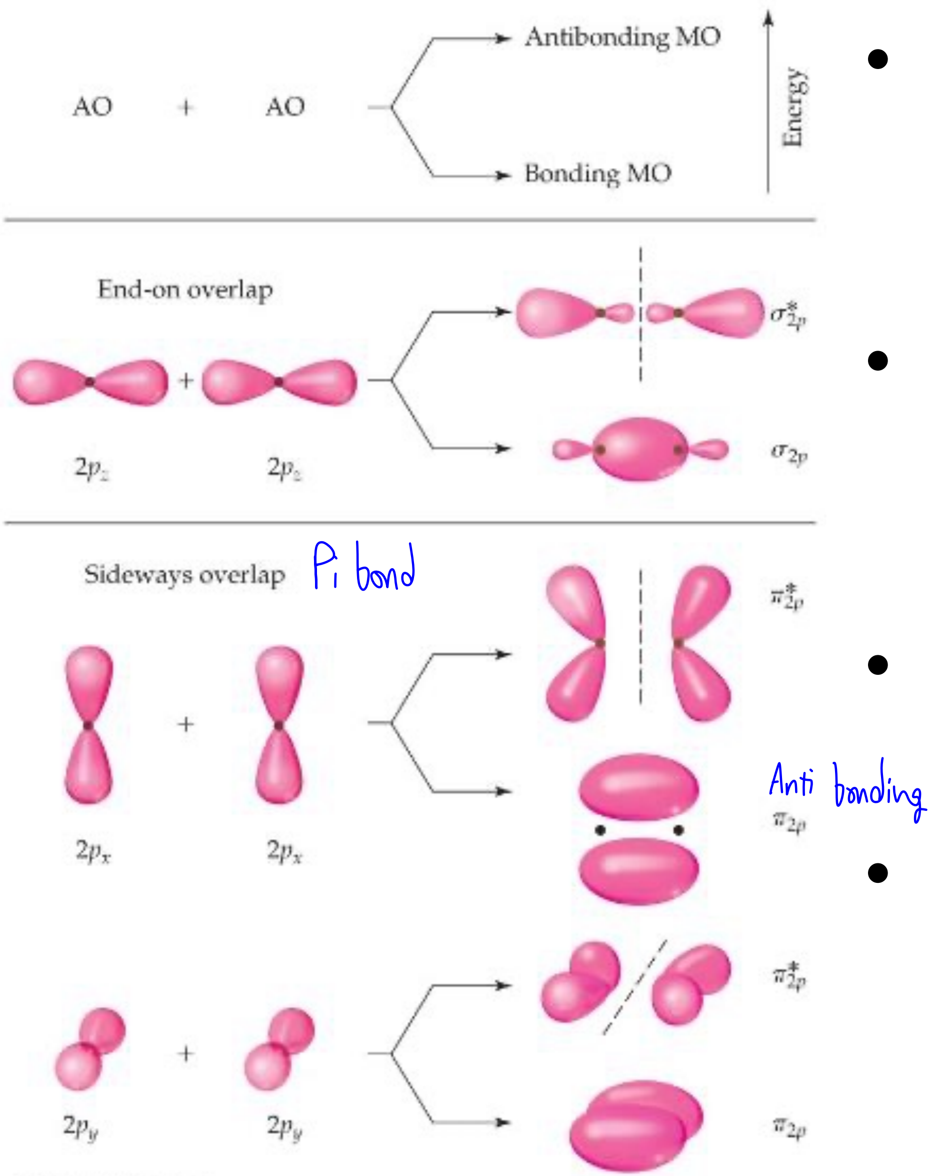
Can He_2 Form? Use MO Diagram and Bond Order to Decide!

- Bond Order = ½(2 – 2) = 0 bonds
- Therefore, He_2 does *not* exist.

No He_2 because not stable



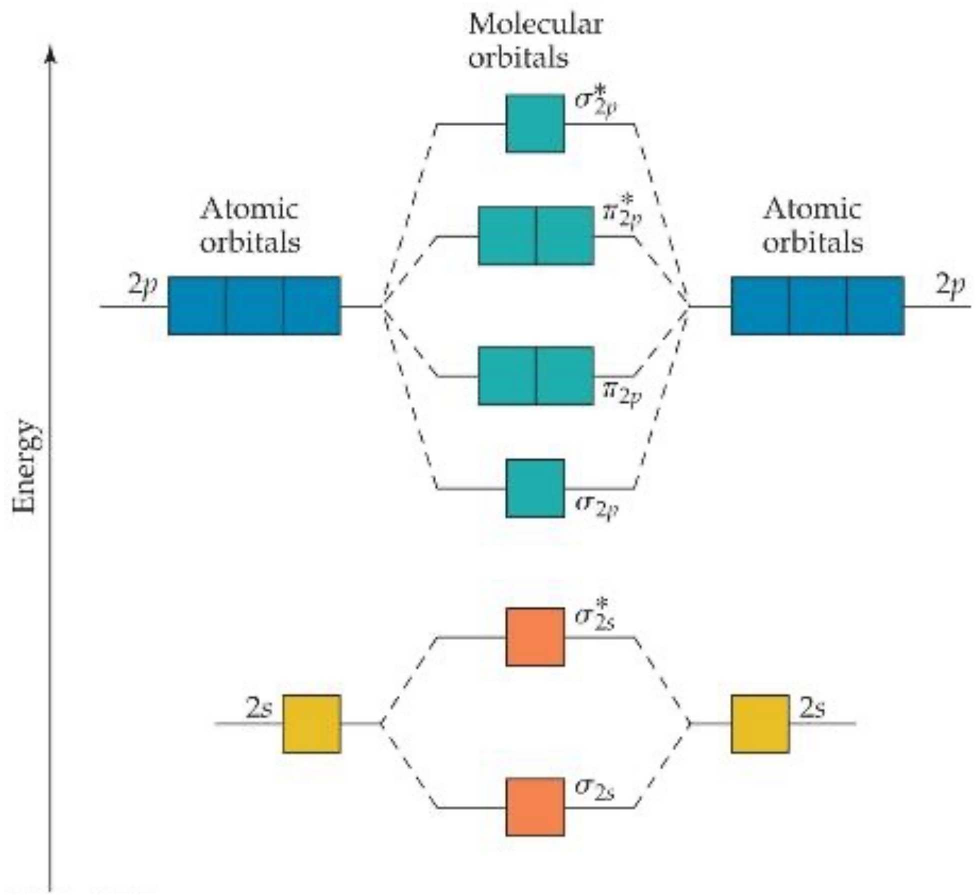
s and *p* Orbitals Can Interact



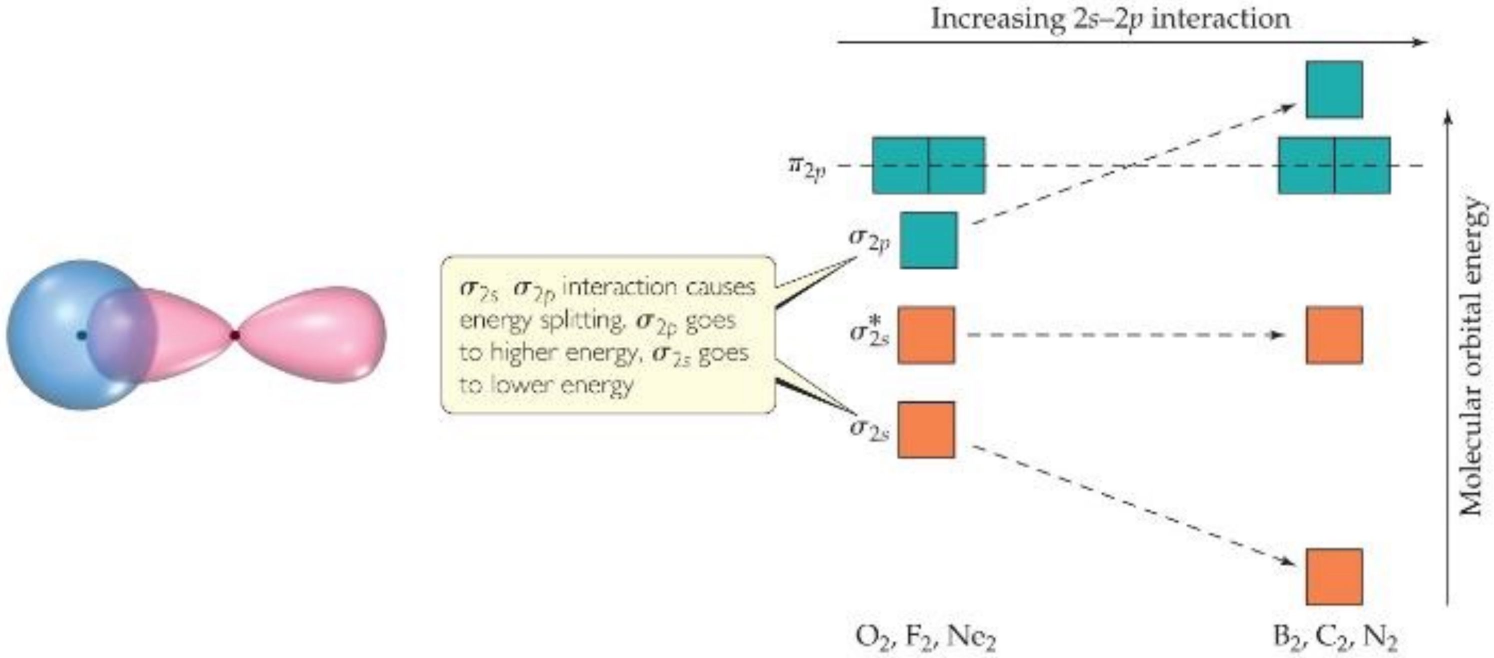
- For atoms with both *s* and *p* orbitals, there are two types of interactions:
- The *s* and the *p* orbitals that face each other overlap in σ fashion.
- The other two sets of *p* orbitals overlap in π fashion.
- These are, again, direct and "side-ways" overlap of orbitals.

MO Theory

- The resulting MO diagram:
  - There are σ and σ^* orbitals from *s* and *p* atomic orbitals.
  - There are π and π^* orbitals from *p* atomic orbitals.
  - Since direct overlap is stronger, the effect of raising and lowering energy is greater for σ and σ^*.



s and p Orbital Interactions



- In some cases, s orbitals can interact wit the p_z orbitals more than the p_x and p_y orbitals.
- It raises the energy of the p_z orbital and lowers the energy of the s orbital.
- The p_x and p_y orbitals are degenerate π_{2p} orbitals.

MO Diagrams for Diatomic Molecules of 2nd Period Elements

Large 2s–2p interaction

	B_2	C_2	N_2
σ^*_{2p}			
π^*_{2p}						
σ_{2p}			↑↓
π_{2p}	↑	↑	↑↓	↑↓	↑↓	↑↓
σ^*_{2s}	↑↓	↑↓	↑↓
σ_{2s}	↑↓	↑↓	↑↓

Small 2s–2p interaction

	O_2	F_2	Ne_2
σ^*_{2p}			↑↓
π^*_{2p}	↑	↑	↑↓	↑↓	↑↓	↑↓
π_{2p}	↑↓	↑↓	↑↓	↑↓	↑↓	↑↓
σ_{2p}	↑↓	↑↓	↑↓
σ^*_{2s}	↑↓	↑↓	↑↓
σ_{2s}	↑↓	↑↓	↑↓

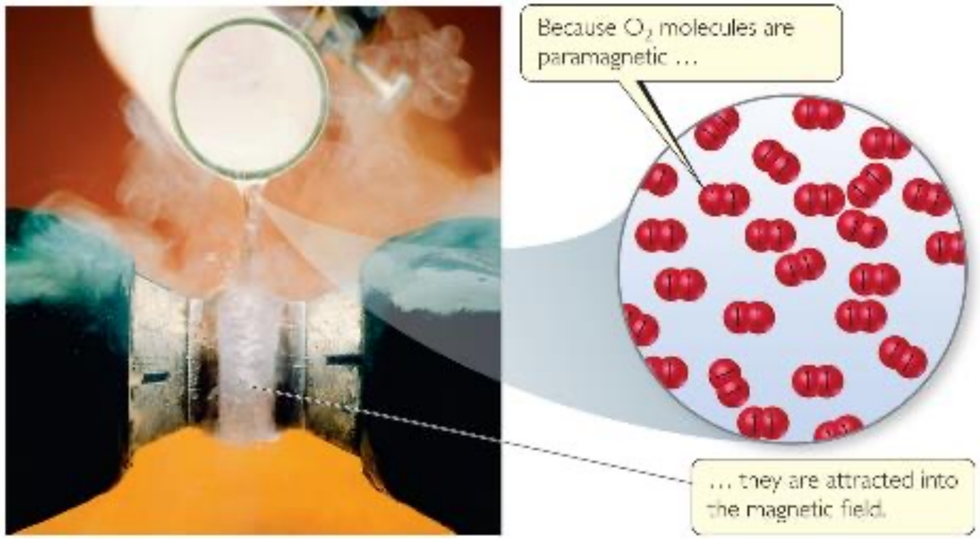
	B_2	C_2	N_2	O_2	F_2	Ne_2
Bond order	1	2	3	2	1	0
Bond enthalpy (kJ/mol)	290	620	941	495	155	—
Bond length (Å)	1.59	1.31	1.10	1.21	1.43	—
Magnetic behavior	Paramagnetic	Diamagnetic	Diamagnetic	Paramagnetic	Diamagnetic	—

MO Diagrams and Magnetism

- **Diamagnetism** is the result of all electrons in every orbital being spin paired. These substances are weakly repelled by a magnetic field.
- **Paramagnetism** is the result of the presence of one or more unpaired electrons in an orbital.
- Is oxygen (O_2) paramagnetic or diamagnetic? Look back at the MO diagram! It is paramagnetic.

Paramagnetism of Oxygen

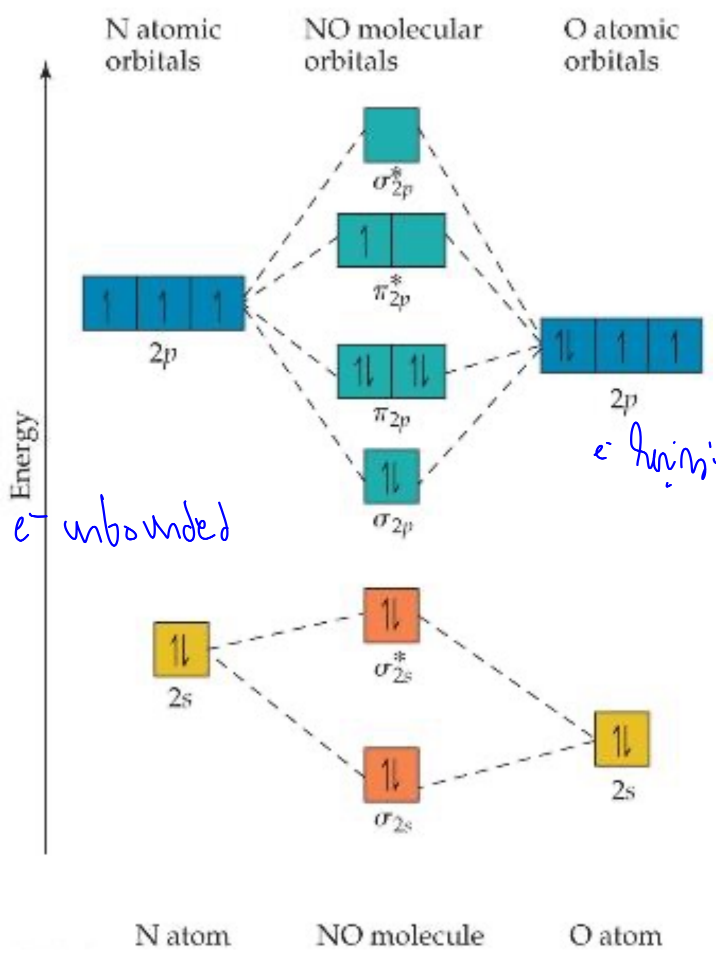
- Lewis structures would *not* predict that O_2 is paramagnetic.
- The MO diagram clearly shows that O_2 is paramagnetic.
- Both show a double bond (bond order = 2).



Heteronuclear Diatomic Molecules

- Diatomic molecules can consist of atoms from different elements.
- How does a MO diagram reflect differences?
- The atomic orbitals have different energy, so the interactions change slightly.
- The more electronegative atom has orbitals lower in energy, so the bonding orbitals will more resemble them in energy.

Paramag: 1e⁻ unbonded

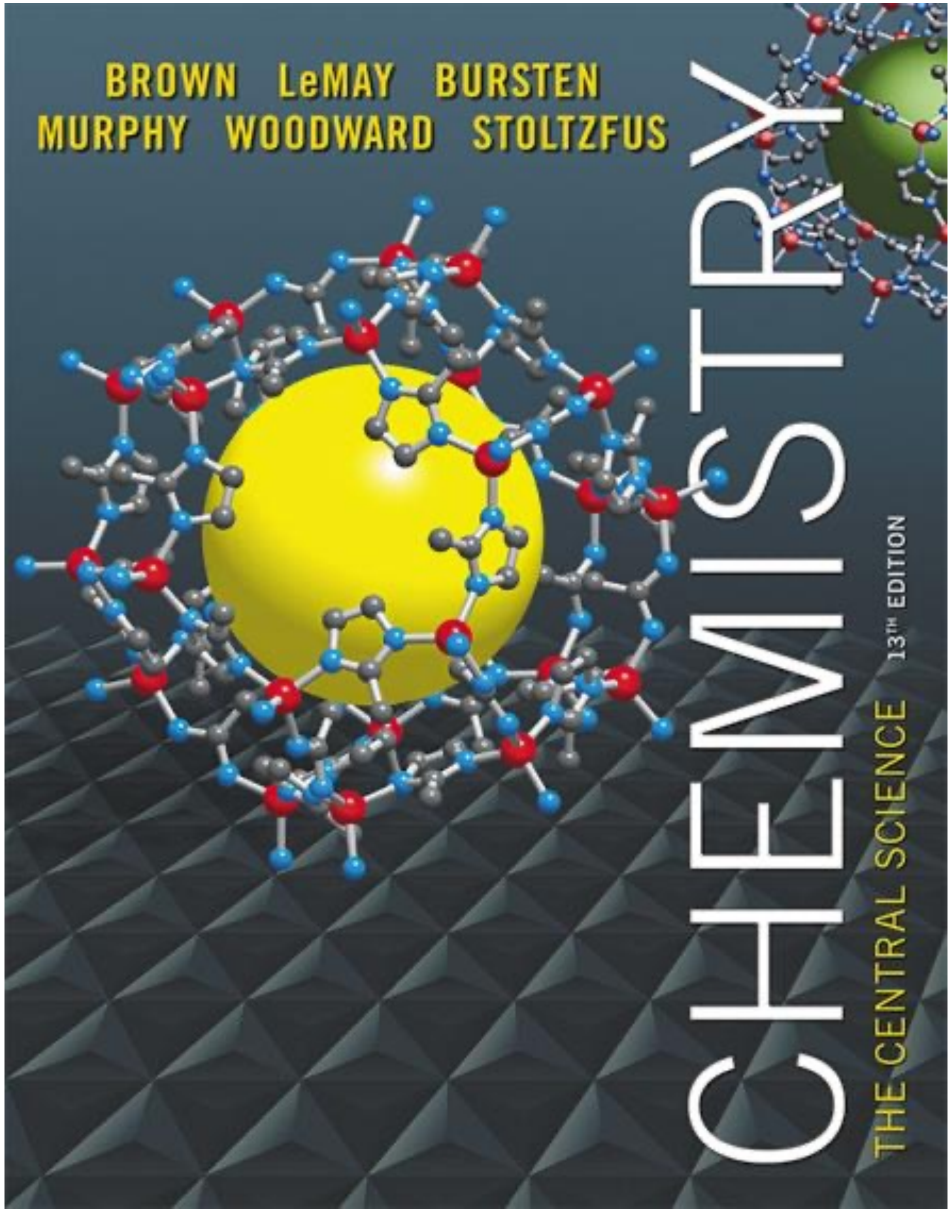


(May not be included in the exam)

Why We Use Both VB Theory and MO Theory

- The exact solution of the Schrödinger equation for real molecules is practically impossible.
- In practice, we use approximation methods to describe molecular systems.
- VB Theory (Valence Bond): uses localized atomic/hybrid orbitals to model bonds as overlaps.
- MO Theory (Molecular Orbital): uses delocalized molecular orbitals with electrons spread over the whole molecule.
- Both are built on the same quantum mechanical foundation; differences come from the choice of basis functions.
- Both are correct and mathematically convertible via unitary transformation.

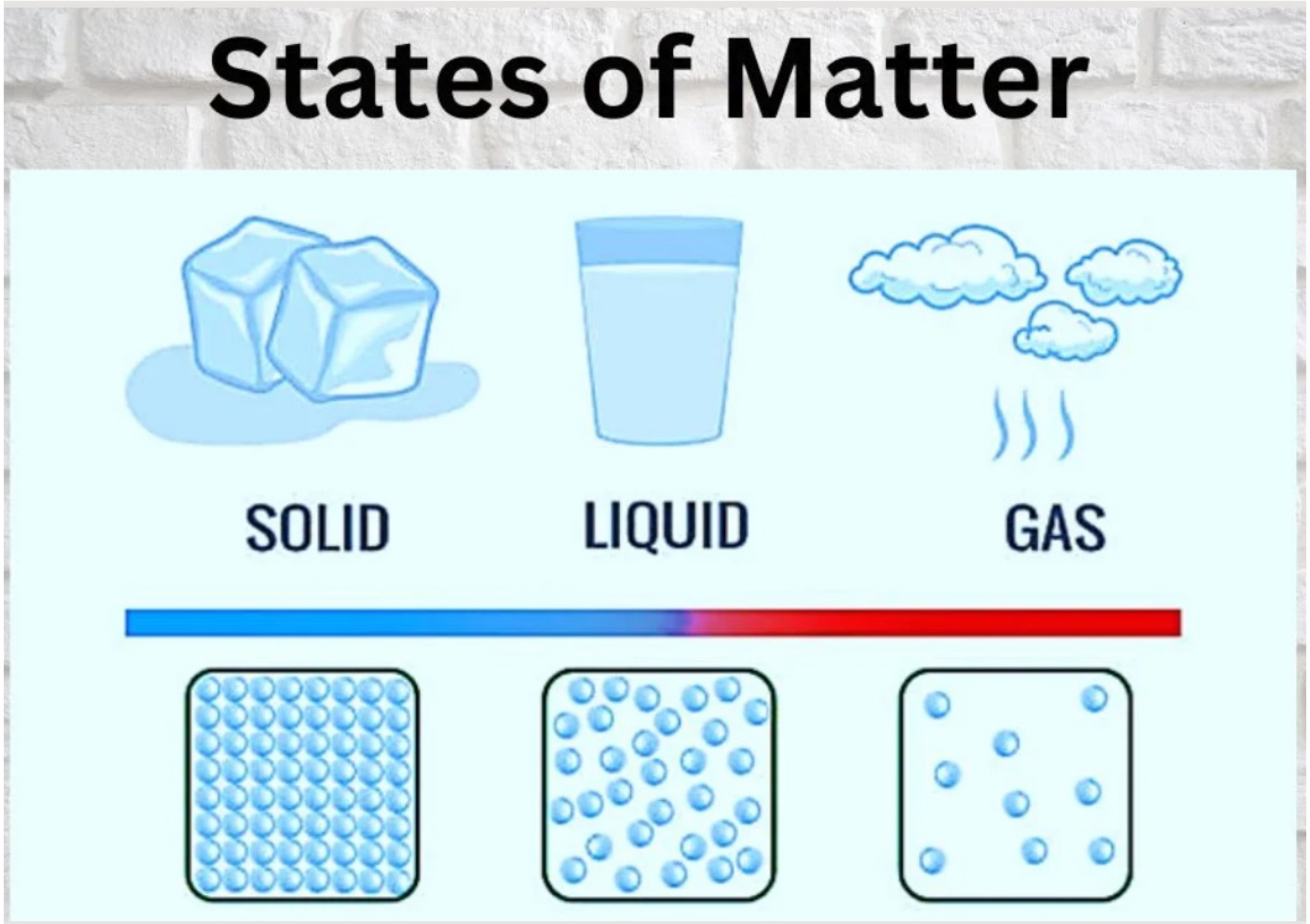
Strengths:
VB → intuitive for shapes, bond angles, Lewis structures.
MO → better for magnetism, delocalization, spectroscopy.
Chemists choose the model according to the problem; using both often gives the most insight.



Lecture Presentation

Chapter 10

Gases


States of Matter
SOLID
LIQUID
GAS

Characteristics of Gases

- Physical properties of gases are all similar.
- Composed mainly of nonmetallic elements with simple formulas and low molar masses.
- Unlike liquids and solids, gases
  - expand to fill their containers.
  - are highly compressible.
  - have extremely low densities.

$PV = nRT$

- Two or more gases form a homogeneous mixture.

Properties Which Define the State of a Gas Sample

1) Temperature
2) Pressure
3) Volume
4) Amount of gas, usually expressed as number of moles

- Having already discussed three of these, we need to define pressure.

Gases

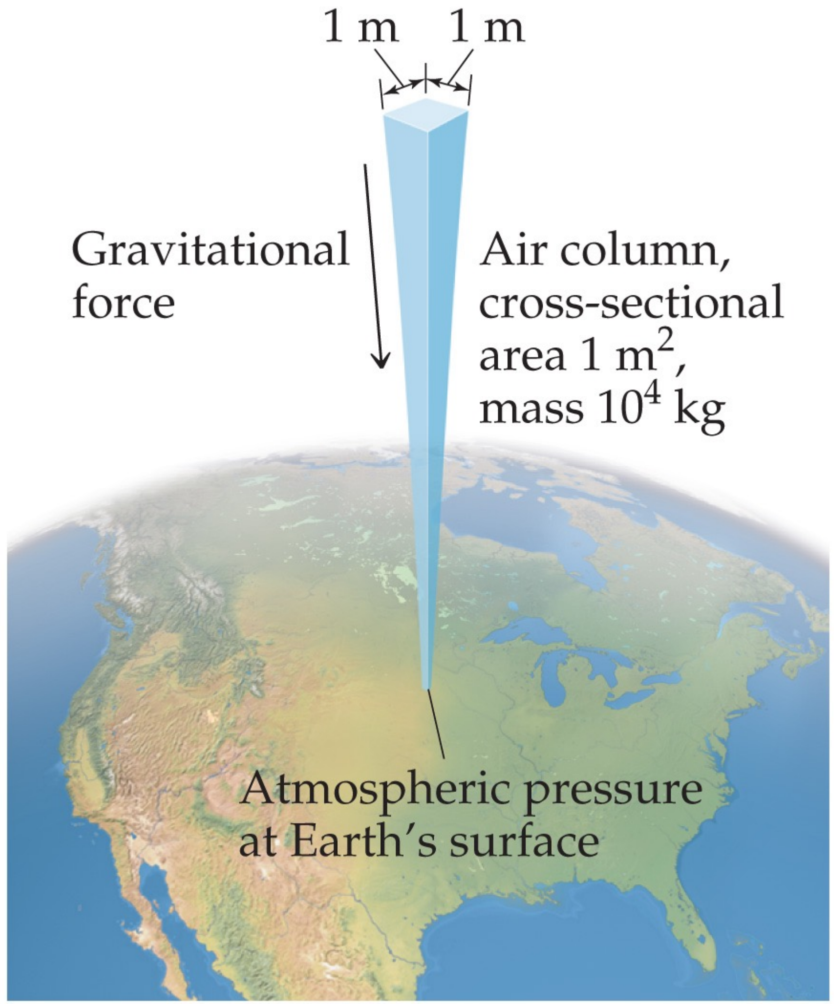
Pressure

- **Pressure** is the amount of force applied to an area:

$$P = \frac{F}{A}$$

- **Atmospheric pressure** is the weight of air per unit of area. ลดถ้าความสูงเพิ่มขึ้น

100°C → boiling point at 1 atm

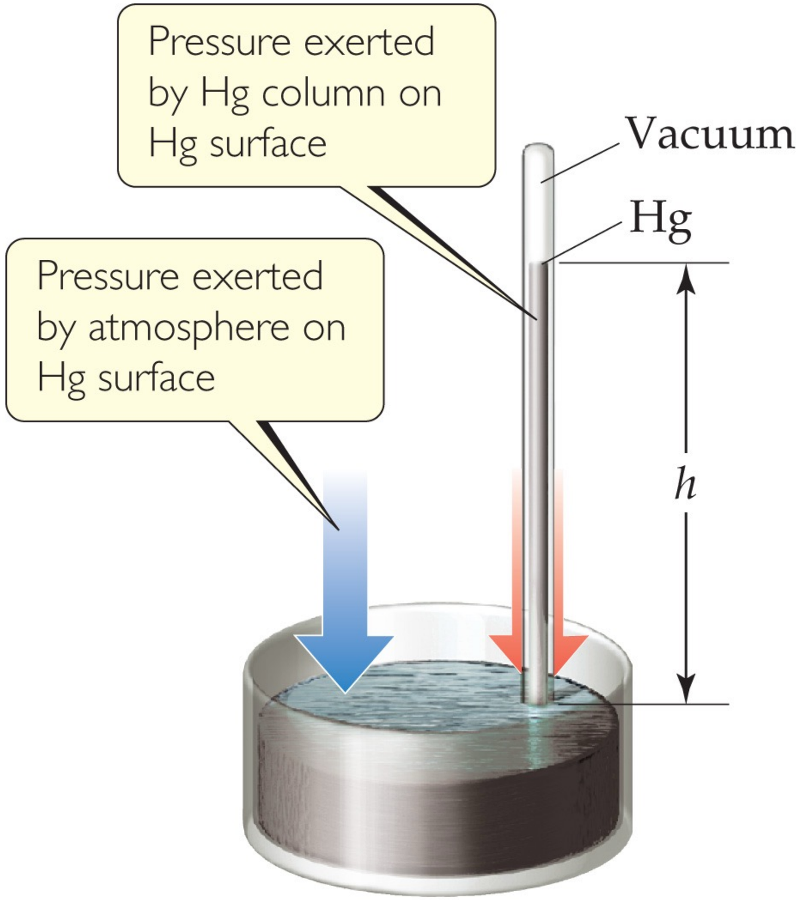


Gases

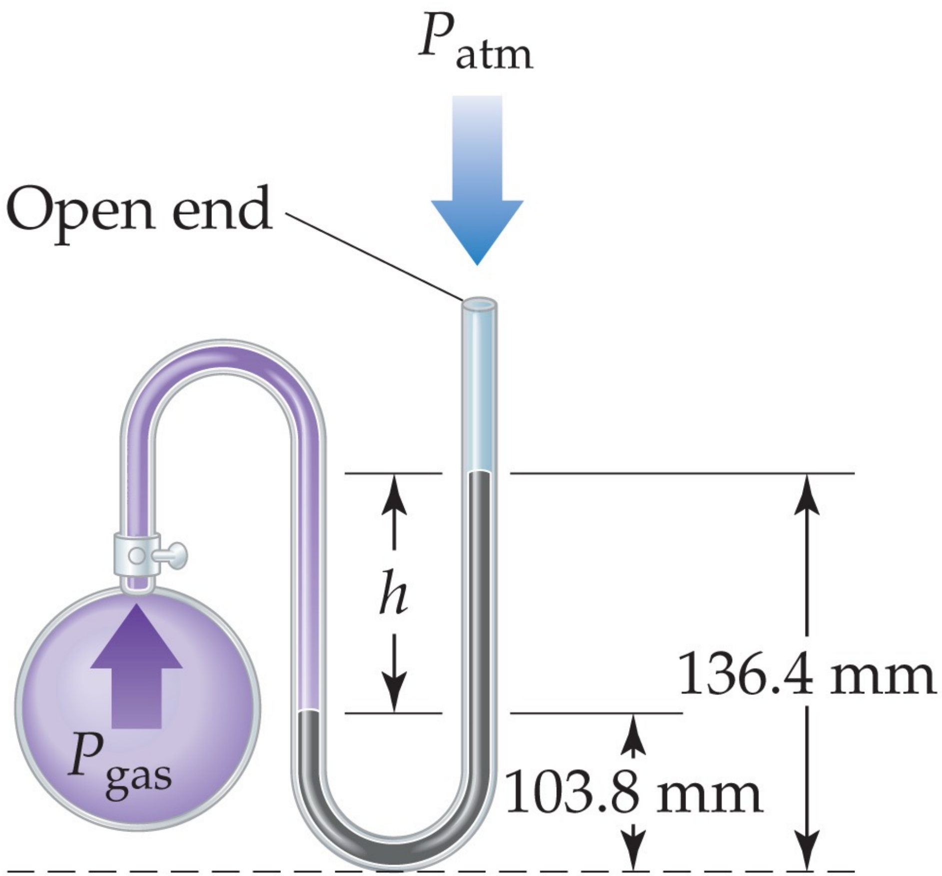
Units of Pressure

- **Pascals**: 1 Pa = 1 N/m^2 (SI unit of pressure)
- **Bar**: 1 bar = 10^5 Pa = 100 kPa
- **mm Hg or torr**: These units are literally the difference in the heights measured in mm of two connected columns of mercury, as in the **barometer** in the figure.
- **Atmosphere**:

1.00 atm = 760 torr = 760 mm Hg = 101.325 kPa



Manometer



$$P_{gas} = P_{atm} + P_h$$

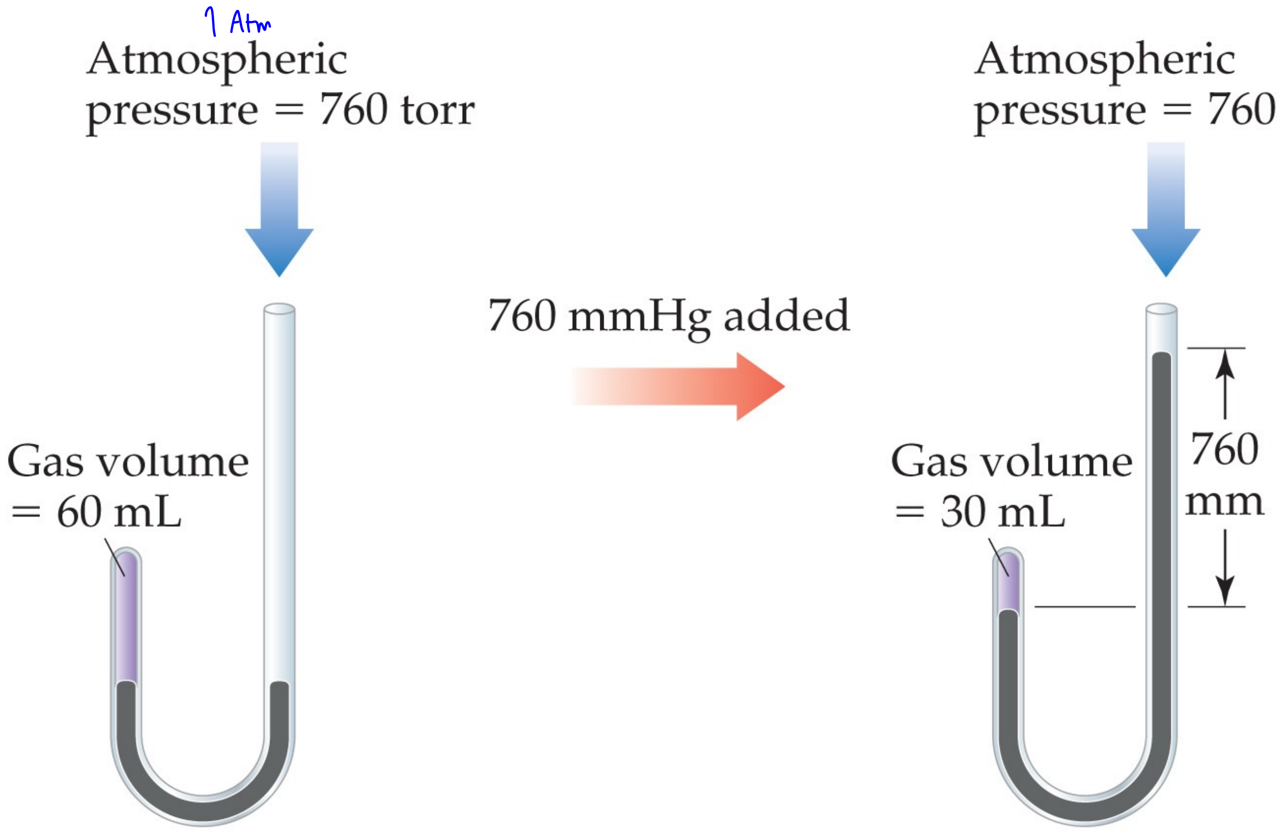
The **manometer** is used to measure the difference in pressure between atmospheric pressure and that of a gas in a vessel. (The barometer seen on the last slide is used to measure the pressure in the atmosphere at any given time.)

Standard Pressure

- Normal atmospheric pressure at sea level is referred to as **standard atmospheric pressure**.
- It is equal to
  - 1.00 atm.
  - 760 torr (760 mmHg).
  - 101.325 kPa.

Boyle's Law

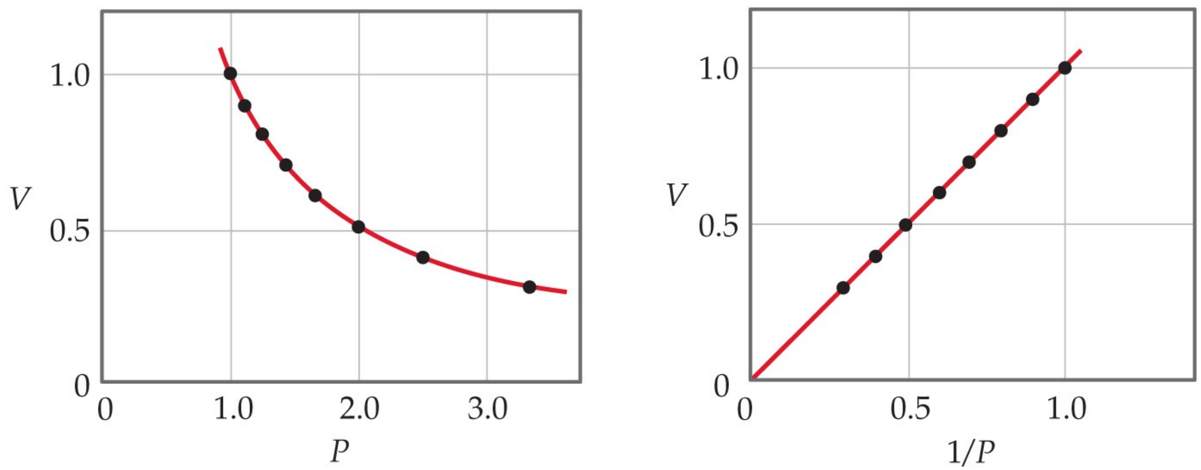
The volume of a fixed quantity of gas at constant temperature is inversely proportional to the pressure.



Gases

Mathematical Relationships of Boyle's Law

- PV = a constant
- This means, if we compare two conditions: $P_1V_1 = P_2V_2$.
- Also, if we make a graph of V vs. P, it will *not* be linear. However, a graph of V vs. $1/P$ will result in a linear relationship!

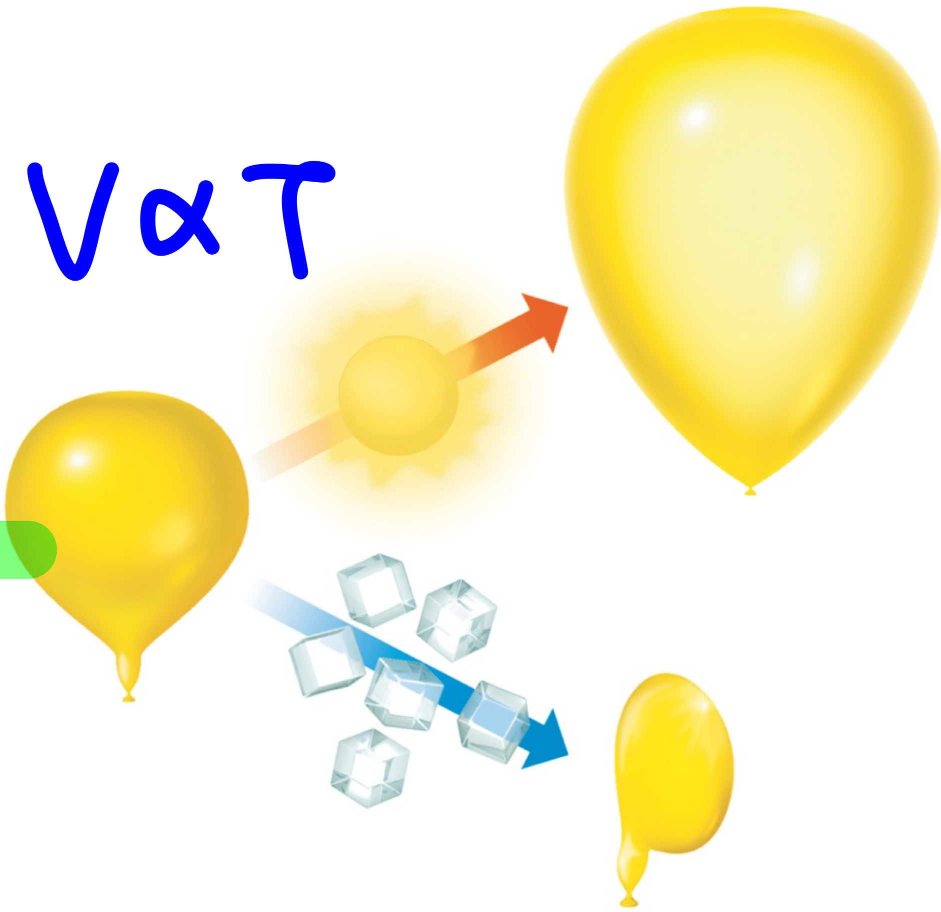


Gases

Charles's Law

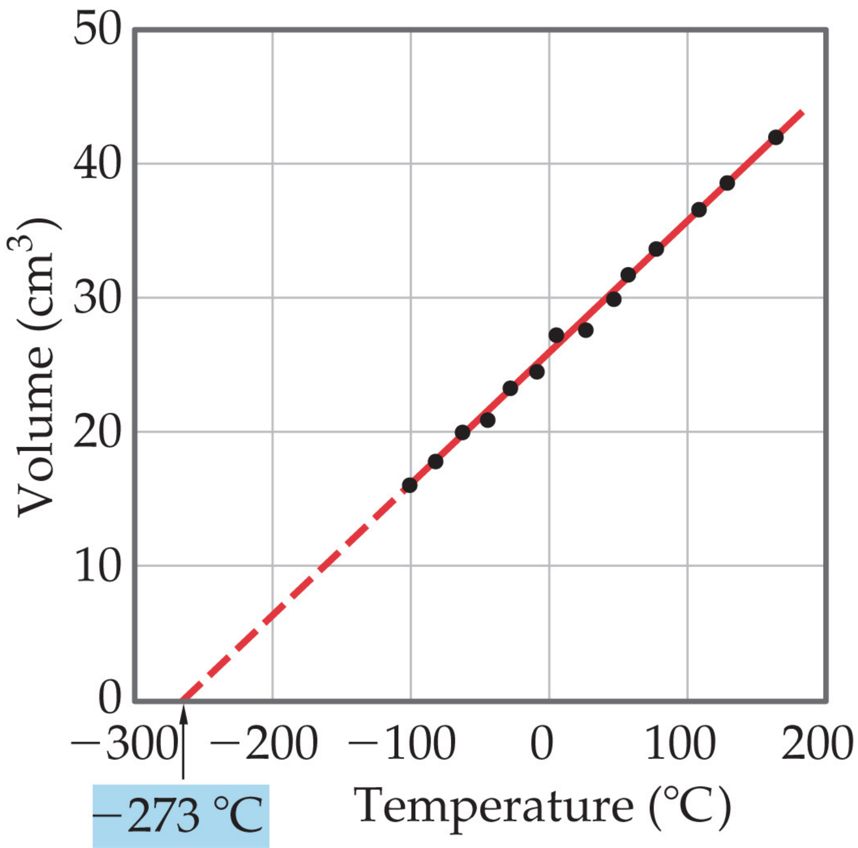
- The volume of a fixed amount of gas at constant pressure is directly proportional to its absolute temperature.

$V \propto T$



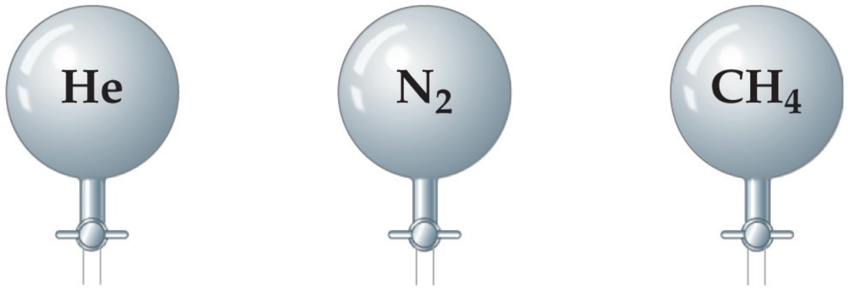
Mathematical Relationships of Charles's Law

- V = constant × T
- This means, if we compare two conditions: $V_1/T_1 = V_2/T_2$.
- Also, if we make a graph of V vs. T, it will be linear.



Avogadro's Law

- The volume of a gas at constant temperature and pressure is directly proportional to the number of moles of the gas.
- Also, at STP, one mole of gas occupies 22.4 L.
- Mathematically: $V = \text{constant} \times n$, or $V_1/n_1 = V_2/n_2$



Volume	22.4 L	22.4 L	22.4 L
Pressure	1 atm	1 atm	1 atm
Temperature	0 °C	0 °C	0 °C
Mass of gas	4.00 g	28.0 g	16.0 g
Number of gas molecules	6.02×10^{23}	6.02×10^{23}	6.02×10^{23}

Ideal-Gas Equation

- So far we've seen that

  $V \propto 1/P$ (Boyle's law).

  $V \propto T$ (Charles's law).

  $V \propto n$ (Avogadro's law).

- Combining these, we get

$$V \propto \frac{nT}{P}$$

- Finally, to make it an equality, we use a constant of proportionality (*R*) and reorganize; this gives the Ideal-Gas Equation: $PV = nRT$.

Density of Gases

If we divide both sides of the ideal-gas equation by *V* and by *RT*, we get

$$n/V = P/RT.$$

Also: moles × molecular mass = mass

$$n \times \mathcal{M} = m.$$

If we multiply both sides by $\mathcal{M}$, we get

$$m/V = \mathcal{M}P/RT$$

and m/V is density, d; the result is:

$$d = \mathcal{M}P/RT.$$

Density & Molar Mass of a Gas

- To recap:
- One needs to know only the molecular mass, the pressure, and the temperature to calculate the density of a gas.
- $d = MP/RT$
- Also, if we know the mass, volume, and temperature of a gas, we can find its molar mass.
- $M = mRT/PV$

Volume and Chemical Reactions

- The balanced equation tells us relative amounts of moles in a reaction, whether the compared materials are products or reactants.
- $PV = nRT$
- So, we can relate volume for gases, as well.
- For example: use ($PV = nRT$) for substance *A* to get moles *A*; use the mole ratio from the balanced equation to get moles *B*; and ($PV = nRT$) for substance *B* to get volume of *B.*

Dalton's Law of Partial Pressures

- If two gases that *don't* react are combined in a container, they act as if they are alone in the container.
- The total pressure of a mixture of gases equals the sum of the pressures that each would exert if it were present alone.
- In other words,

$$P_{\text{total}} = p_1 + p_2 + p_3 + \ldots$$

Mole Fraction

- Because each gas in a mixture acts as if it is alone, we can relate amount in a mixture to partial pressures:

$$\frac{P_1}{P_t} = \frac{n_1 RT/V}{n_t RT/V} = \frac{n_1}{n_t}$$

- That ratio of moles of a substance to total moles is called the **mole fraction**, χ.

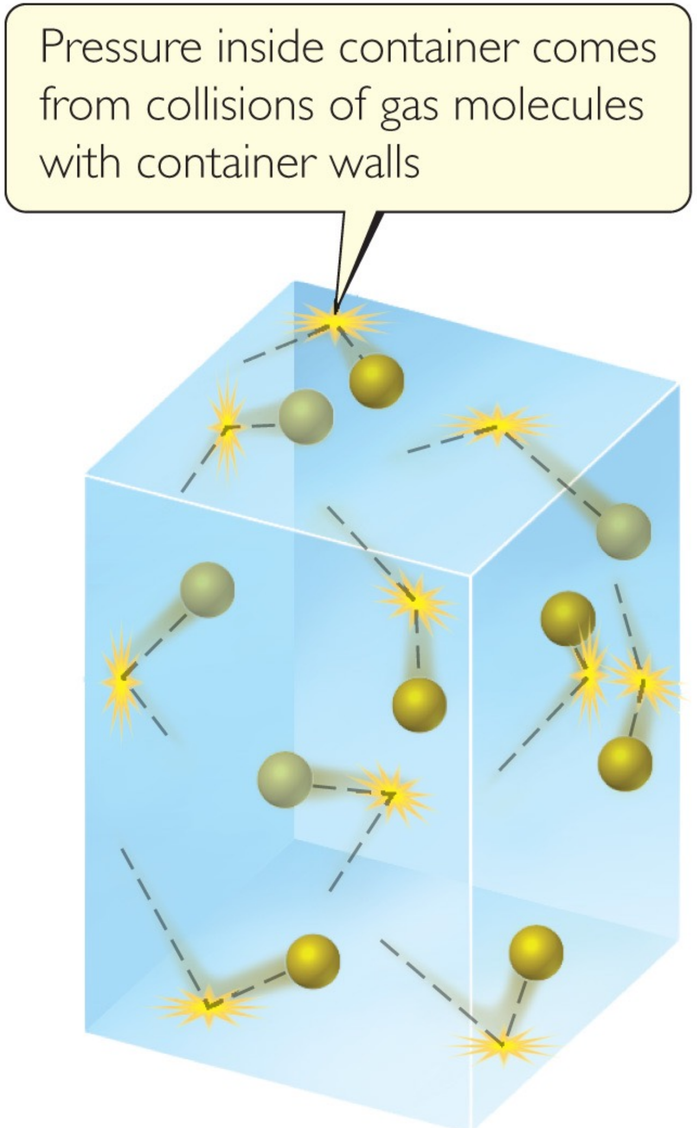
$$X_1 = \frac{\text{Moles of compound 1}}{\text{Total moles}} = \frac{n_1}{n_t}$$

Pressure and Mole Fraction

- The end result is

$$P_1 = \left(\frac{n_1}{n_t}\right)P_t = X_1 P_t$$

Kinetic-Molecular Theory



- *Laws* tell us *what* happens in nature. Each of the gas laws we have discussed tell us what is observed under certain conditions.
- *Why* are these laws observed? We will discuss a *theory* to explain our observations.

Main Tenets of Kinetic-Molecular Theory

1) Gases consist of large numbers of molecules that are in continuous, random motion.

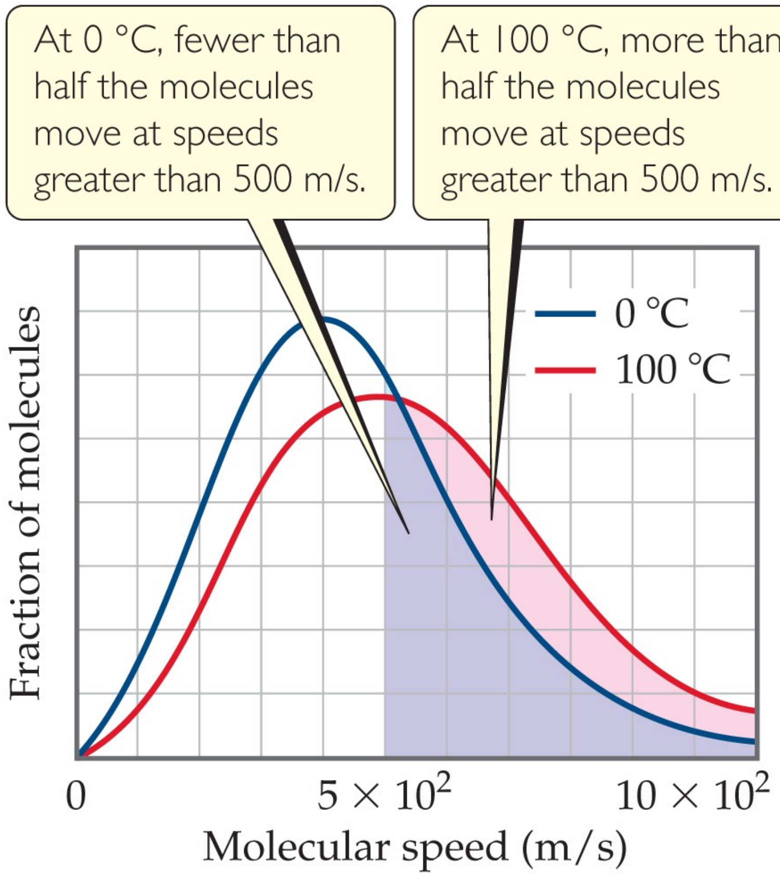
large number of molecules

2) The combined volume of all the molecules of the gas is negligible relative to the total volume in which the gas is contained.

3) Attractive and repulsive forces between gas molecules are negligible.

Gases

Main Tenets of Kinetic-Molecular Theory



(a)

4) Energy can be transferred between molecules during collisions, but the *average* kinetic energy of the molecules does not change with time, as long as the temperature of the gas remains constant.
5) The average kinetic energy of the molecules is proportional to the absolute temperature.

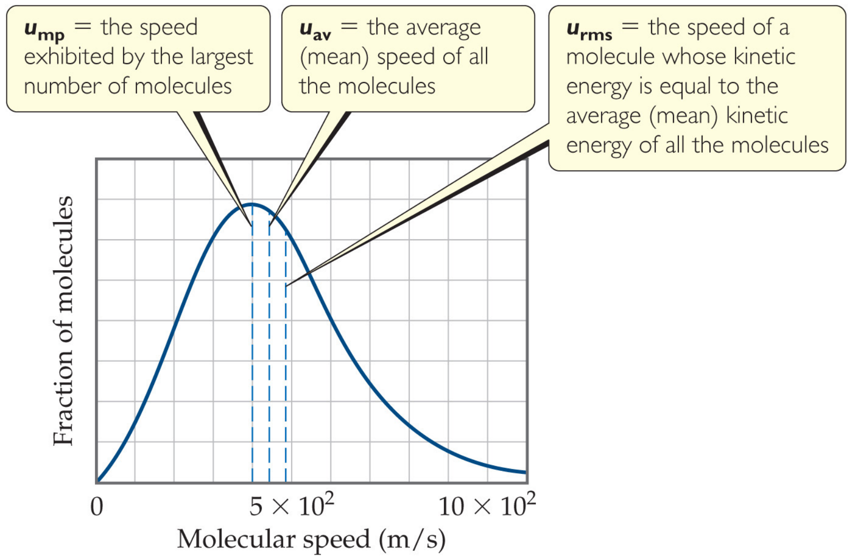
How Fast Do Gas Molecules Move?

- Temperature is related to their *average* kinetic.
- Individual molecules can have different speeds of motion.
- The figure shows three different speeds:

➢ u_{mp} is the most probable speed (most molecules are this fast).

➢ u_{av} is the average speed of the molecules.

➢ u_{rms}, the root-mean-square speed, is the one associated with their average kinetic energy.



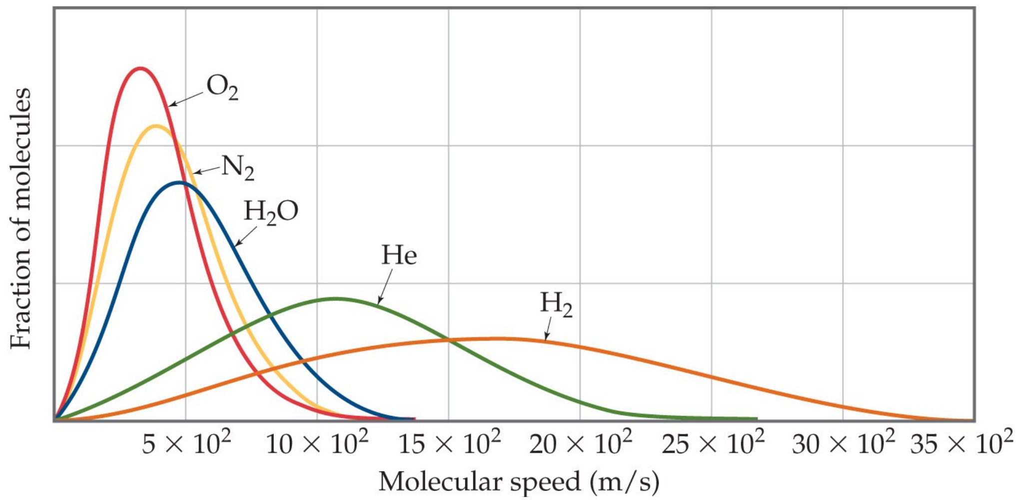
(b)

Gases

u_{rms} and Molecular Mass

$$u_{rms} = \sqrt{\frac{3RT}{\mathcal{M}}}$$

- At any given temperature, the average kinetic energy of molecules is the same.
- So, ½ m $(u_{rms})^2$ is the same for two gases at the same temperature.
- If a gas has a low mass, its speed will be greater than for a heavier molecule.



- The empirical observations of gas properties as expressed by the various gas laws are readily understood in terms of the kinetic-molecular theory. The following examples illustrate this point:

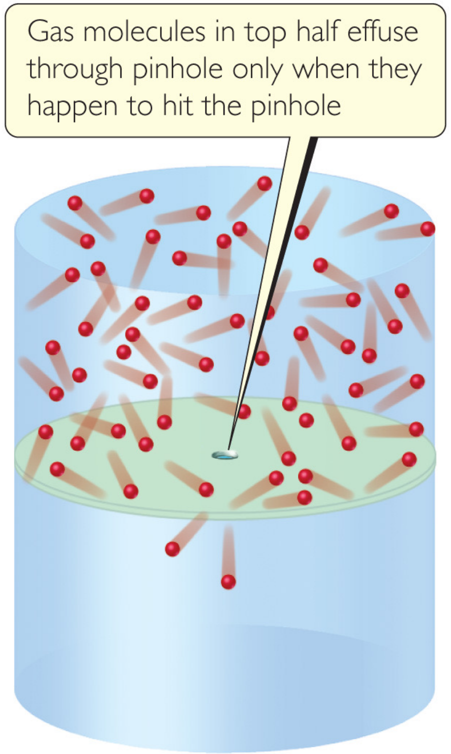
- **1. An increase in volume at constant temperature causes pressure to decrease.** A constant temperature means that the average kinetic energy of the gas molecules remains unchanged. This means that the rms speed of the molecules remains unchanged. When the volume is increased, the molecules must move a longer distance between collisions. Consequently, there are fewer collisions per unit time with the container walls, which means the pressure decreases. Thus, kinetic- molecular theory explains Boyle's law.
- **2. A temperature increase at constant volume causes pressure to increase.** An increase in temperature means an increase in the average kinetic energy of the molecules and in u_{rms}. Because there is no change in volume, the temperature increase causes more collisions with the walls per unit time because the molecules are all moving faster. Furthermore, the momentum in each collision increases (the molecules strike the walls more forcefully). A greater number of more forceful collisions means the pressure increases, and the theory explains this increase.

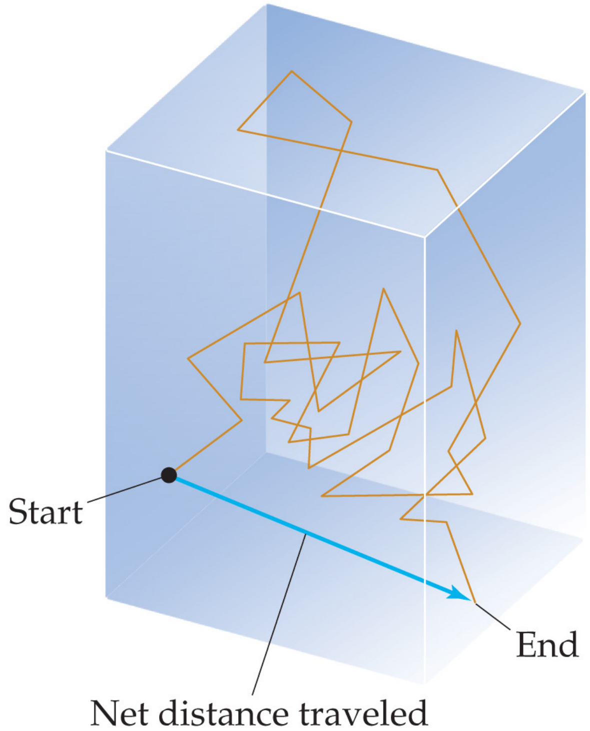
Effusion & Diffusion

depend on molecular motion

Effusion is the escape of gas molecules through a tiny hole into an evacuated space.

Diffusion is the spread of one substance throughout a space or a second substance.





Gases

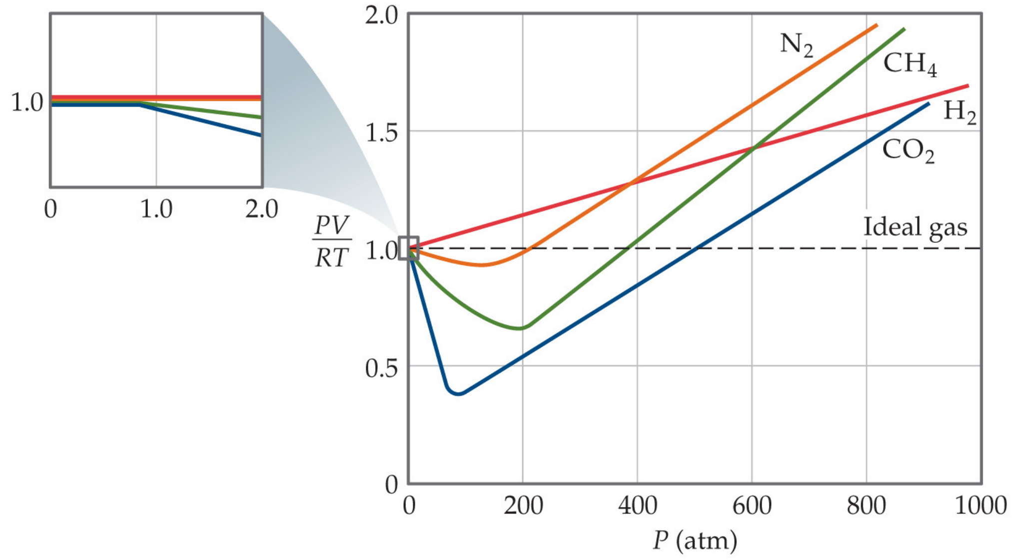
Graham's Law Describes Diffusion & Effusion

- Graham's Law relates the molar mass of two gases to their rate of speed of travel.
- The "lighter" gas always has a faster rate of speed.

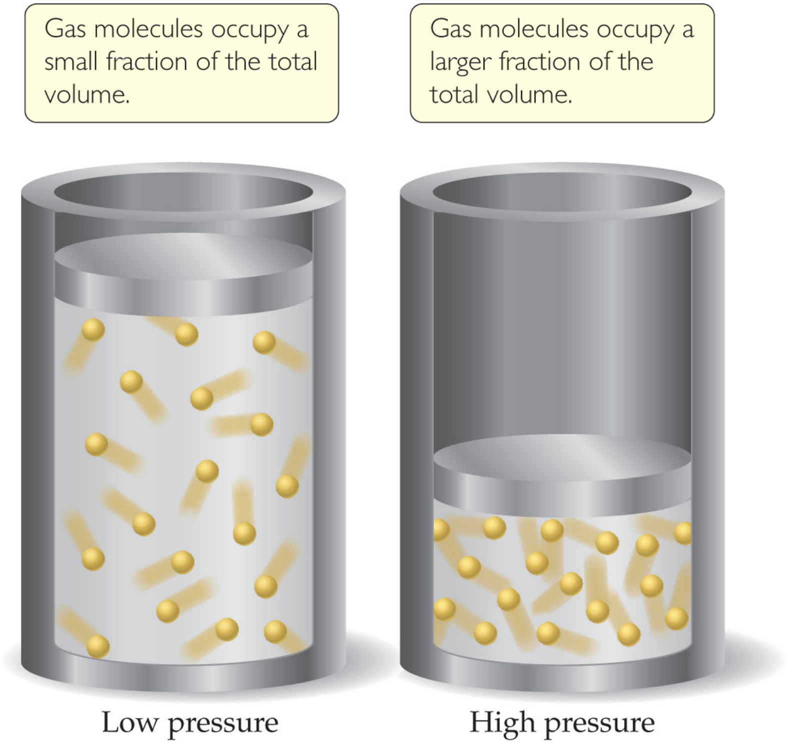
$$\frac{r_1}{r_2} = \sqrt{\frac{\mathcal{M}_2}{\mathcal{M}_1}}$$

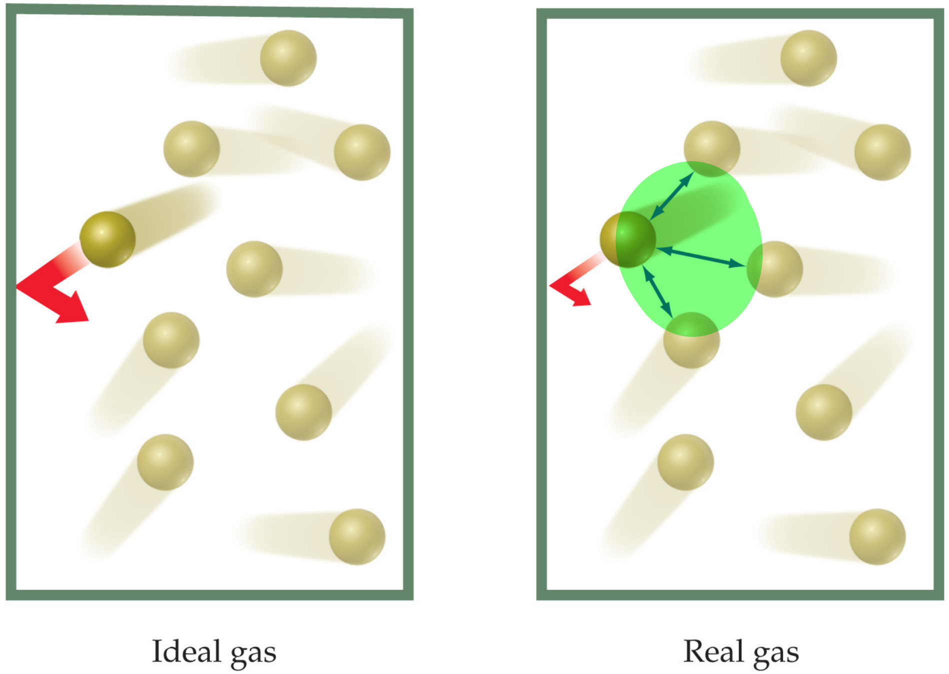
Real Gases

- In the real world, the behavior of gases only conforms to the ideal-gas equation at relatively high temperature and low pressure.
- Even the same gas will show wildly different behavior under high pressure at different temperatures.



Deviations from Ideal Behavior





The assumptions made in the kinetic-molecular model (negligible volume of gas molecules themselves, no attractive forces between gas molecules, etc.) break down at high pressure and/or low temperature.

Corrections for Nonideal Behavior

- The ideal-gas equation can be adjusted to take these deviations from ideal behavior into account.
- The corrected ideal-gas equation is known as the **van der Waals equation**.
- The pressure adjustment is due to the fact that molecules attract and repel each other.
- The volume adjustment is due to the fact that molecules occupy some space on their own.

The van der Waals Equation

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

Table 10.3 Van der Waals Constants for Gas Molecules

Substance	a (L^2-atm/mol^2)	b (L/mol)
He	0.0341	0.02370
Ne	0.211	0.0171
Ar	1.34	0.0322
Kr	2.32	0.0398
Xe	4.19	0.0510
H_2	0.244	0.0266
N_2	1.39	0.0391
O_2	1.36	0.0318
F_2	1.06	0.0290
Cl_2	6.49	0.0562
H_2O	5.46	0.0305
NH_3	4.17	0.0371
CH_4	2.25	0.0428
CO_2	3.59	0.0427
CCl_4	20.4	0.1383